

COMPUTER SCIENCE TRIPOS Part IB, Part II 50% – 2022 – Paper 7

3 Economics, Law and Ethics (ah793)

You lead the security team for a large webmail provider.

Behavioural
economics

- (a) You are seeking additional resources to boost your team's capabilities. How might you frame your request for an increased budget to increase the likelihood it will be approved? [8 marks]

Answer: According to the decision-making process of bounded rationality, framing actions as savings can make them more attractive (2 marks). Therefore, the focus should be on how much money the company can save by improving the security team, rather than the security problems that are likely if the team is not strengthened (2 marks). Furthermore, you may want to focus on immediate savings (2 marks). People are likely to disregard events that are likely to happen in the distant future, compared to more immediate events (hyperbolic discounting) (2 marks).

- (b) A crime group is using accounts with the webmail provider to organise the importation of firearms into the UK. The police would like to know more about their operations.

Law and the
Internet

- (i) How might the police discover more about the plans? [4 marks]

Answer: The police can arrange a warrant for the targeted interception of the communications (2 marks) under the Investigatory Powers Act 2016 (2 marks).

Philosophies of
ethics

- (ii) Apply ethical philosophies to argue for and against your suggested approach. [8 marks]

Answer: Consequentialism focuses on the end rather than the means (2 mark), so targeted interception would be justified if it were to result in just consequences (e.g. disruption to serious and organised crime, terrorism) (2 marks). The Rawlsian approach says that we should maximise utility for those worst off (2 mark). In relation to targeted interception, there is a risk for misuse for political dissidents and minorities, and risks associated with the misuse of the powers, either by the police, or if the data were to fall into the wrong hands (2 marks).
